Motor Unit Firing Properties During Maximal, Dynamic Muscle Actions

CHRISTOPHER A. RIVAS, CALEB VOSKUIL, MONIQUE DUDAR, ALISA SOLOVEVA, AND DR. JOSHUA C. CARR

Department of Kinesiology, Texas Christian University, Fort Worth, TX, USA

Christopher A. Rivas
TCU Harris College of Nursing and Health Sciences
Neuromuscular Physiology Laboratory



BACKGROUND

Each motor unit (MU) possesses inherent firing and recruitment properties that can be captured with surface electromyography (EMG) techniques. Surface EMG decomposition is an emerging technique that allows for the motor unit pools' firing properties to be quantified. Previous literature has observed these properties during isometric muscle actions, however, recent advancements in artificial intelligence for EMG decomposition algorithms have allowed the potential to capture firing properties during dynamic muscle actions.

Research Question:

Using the Precision Decomposition III (PDIII) algorithm, can changes in motor unit firing properties be captured during maximal effort muscle actions of varying load?

METHODS

Participants:

- 23 (8M & 15F) college-aged participants
- 11 Trained, 12 Untrained

Assessment:

Bicep Curl One Repetition Maximum Strength Testing:

- Maximum of 5 attempts
- A heavier repetition is defined as the attempt in which the true 1 repetition maximum was reached and the lighter repetition was a successful attempt prior to the true 1 repetition maximum

Surface Electromyography:

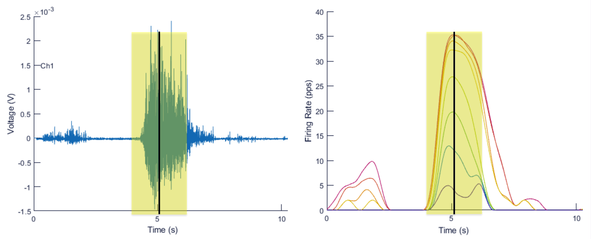
- Collected on the testing arm
- EMG sensor placed on bicep brachii at 66% length of the humerus

Analysis:

Using Delsys NeuroMap software, surface EMG signals were decomposed into their respective motor unit trains. Repetitions were analyzed only when ≥ 5 motor units were above an 80% confidence level. A 2-second interval was used to gather the firing properties of individual motor units. From this interval, motor unit activity was quantified with the below outcomes:

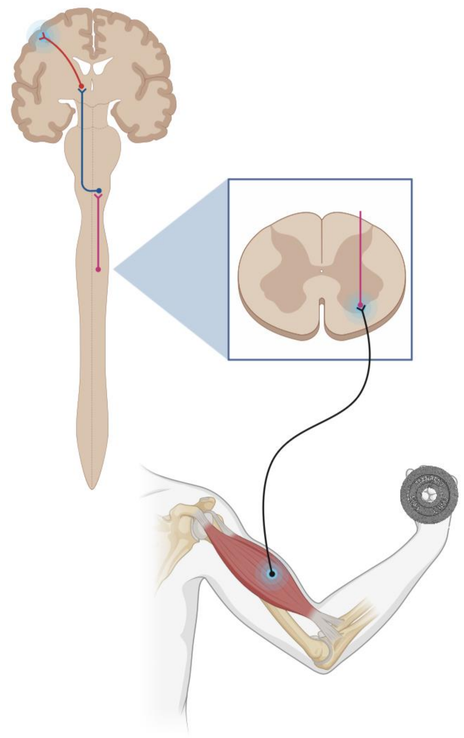
- Total number of motor units
- Peak and average firing rate
- Peak and average motor unit action potential
- Firing rate and motor unit action potential relationship and the slope and y-intercept of this relationship

Paired-sample t-tests were used to determine if there was a statistical difference between heavier and lighter repetitions.

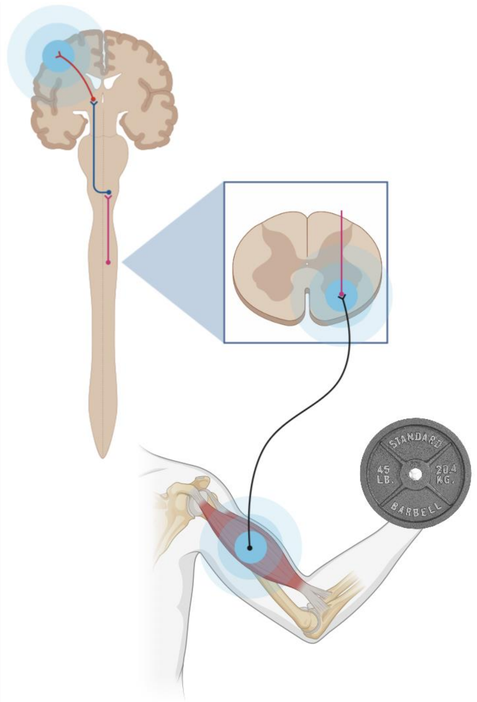


RESULTS

Lighter Load



Heavier Load



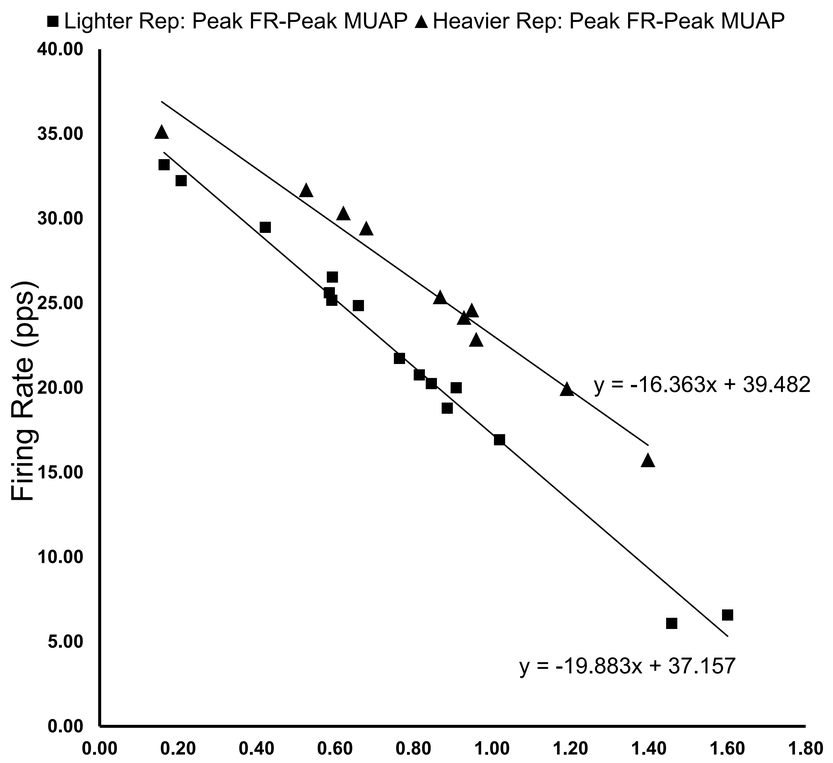


Figure 1. An example of the changes in motor unit firing properties from one participant during maximal effort bicep curls at the '"lighter" 29 lbs. and "heavier" 31.5 lbs. loads

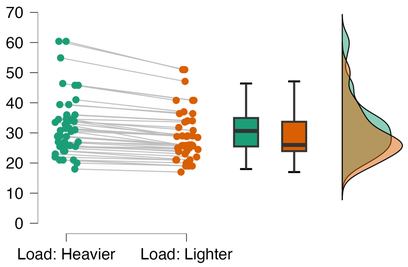


Figure 2. Individual load (lbs.) values of heavier and lighter repetitions

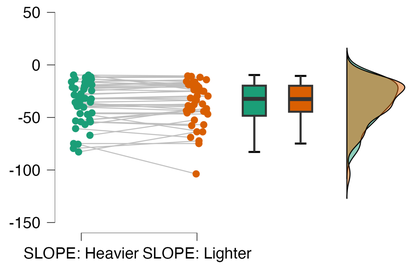


Figure 3. Individual slope values of heavier and lighter repetitions

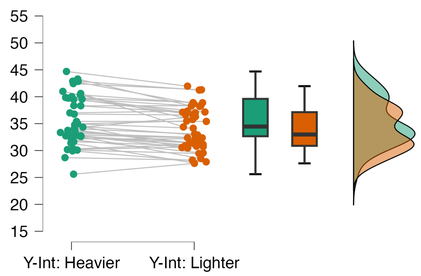


Figure 4. Individual y-intercept (pps) values of heavier and lighter repetitions

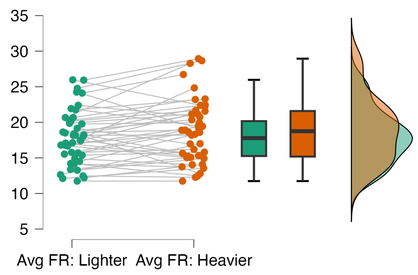


Figure 5. Individual average firing rate (pps) values of heavier and lighter repetitions

LEGEND

Slope: The relationship between firing rate and action potential

Y-intercept: The maximum firing rate able to be achieved

Firing Rate: The rate at which the α-Motor neuron sends action potentials to the muscle fibers it innervates

Motor Unit Action Potential: The sum of the action potentials generated by a motor unit

RESULTS

Variable	Lighter Rep Mean (SD)	Heavier Rep Mean (SD)	p-value	Effect Size
Load (lbs.)	29.06 (8.29)	32.12 (10.02)	<0.001	1.000
Slope	-36.08 (20.43)	-36.01 (20.21)	0.535	-0.014
Y-int	33.91 (4.01)	35.51 (4.5)	<0.001	0.884
Peak FR (pps)	21.36 (3.94)	21.67 (4.30)	0.097	0.231
Avg FR (pps)	17.89 (3.80)	18.84 (4.50)	0.016	0.334
Peak MUAP (mV)	0.47 (0.29)	0.52 (0.37)	0.047	0.298
Avg MUAP (mV)	0.37 (0.22)	0.43 (0.30)	0.007	0.450
Number	8.77 (3.27)	9.71 (3.42)	0.050	0.252

Table 1. Descriptives and significance values of MU firing property comparisons

RESULTS

- The heavier attempt displayed increases in average firing rate, peak motor unit action potential, and average motor unit action potential
- There was no significant difference between heavier and lighter repetitions for slope, peak firing rate, and number
- Load, slope, peak firing rate, peak motor unit action potential, and avg motor unit action potential violated assumptions of normality, so these were corrected using a Wilcoxon signed-rank test

SUMMARY STATEMENTS

- We show that on average, a modest increase of about 3 lbs. in load resulted in significant increases in motor unit firing behavior during maximal effort bicep curls.
- Our finding of significant increase in average firing rate ($p=0.016$) and y-intercept ($p<0.001$), with no difference in slope ($p=0.535$), demonstrates that the firing rate of the entire motor unit pool increases to meet task demands, i.e. heavier loads.
- These findings provide confidence that the PDIII algorithm is capable of capturing motor unit firing properties during maximal effort muscle actions of varying load due to the observed recruitment patterns. These patterns match what is expected based on the physiological principle known as the "Hennemann's Size Principle".